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Roll No. _____ **Total No. of Pages : 03**

Total No. of Questions : 18

BBA (Sem.–5)

OPERATION RESEARCH

Subject Code : BBA-501-18

M.Code : 78193

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTIONS-B consists of FOUR Sub-sections : Units-I, II, III & IV.**
3. **Each Sub-section contains TWO questions each, carrying TEN marks each.**
4. **Student has to attempt any ONE question from each Sub-section.**

SECTION-A

1) Define OR and discuss its scope.

2) Define the simplex method.

3) Define stepping stone method.

4) What is a Travelling Salesman Problem?

5) Define assignment problem.

6) Define slack and float.

7) Explain the terms: Individual, preventive and group replacements.

8) How to compute the Reorder Level of inventory?

9) Define sequencing problem.

10) Define inventory cost.

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SECTION-B

UNIT-I

11) What is problem formulation in operation research? State and describe the various components of the problem formulation.

12) Give a general mathematical formulation of the L.P. problem. Explain all terms.

UNIT-II

13) What do you understand by the initial feasible solution? Explain the concept of North - West Corner rule/the Least Cost Method/VAM method.

14) What is an assignment problem? Describe the mathematical formulation of an assignment problem.

UNIT-III

15) A small project consisting of eight activities has the following characteristics :

Activity	Preceding Activity	Most Optimistic Time (a)	Most Likely Time (m)	Most Pessimistic Time (b)
A	None	2	4	12
B	None	10	12	26
C	A	8	9	10
D	A	10	15	20
E	A	7	7.5	11
F	B, C	9	9	9
G	D	3	3.5	7
H	E, F, G	5	5	5

a) Draw the PERT network for the project.

b) Prepare the activity schedule for the project.

c) Determine the critical path.

16) a) Difference between PERT and CPM. Under what circumstances would you consider PERT as opposed to CPM in project management?

b) Discuss how to process n jobs through m machines.

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UNIT-IV

17) Derive the condition for the optimal replacement age of items whose maintenance costs increase with time, assuming that money value remains constant.

18) A manufacturer has to supply 12,000 units of a product per year to his customer. The demand is fixed and known and the shortage cost is assumed is to be infinite. The inventory holding cost is Rs. 0.20 per unit per month and the setup cost per run is Rs. 350. Determine :

a) The optimum run size q_0

b) Optimum scheduling period t_0

c) Minimum total variable yearly cost.

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